INVERTIS UNIVERSITY, BAREILLY

B.Tech CSE – Third Year | DBMS & SQL Lab (BCS552)

EXPERIMENT NO. 5

1. Experiment Title:

Pattern Matching with LIKE

2. Aim:

Find book titles that match specified text patterns using the LIKE operator.

3. Software / Tools Required:

SQL database system (such as MySQL, PostgreSQL, or SQLite where supported), SQL editor/IDE, and a computer.

4. Theory:

LIKE is used with WHERE for pattern matching. The percent sign (%) matches any sequence of characters, including no characters. The underscore (_) matches exactly one character.

5. Algorithm / Procedure:

1. Create the book table and insert the four book records.
2. Use 'Data%' to find titles starting with Data.
3. Use '%C' to find titles ending with C.
4. Use '_h%' to find titles whose second character is h.

6. Program / Queries / Commands:

```
CREATE TABLE book (
    title VARCHAR(40),
    category VARCHAR(20),
    copies INT
);

INSERT INTO book (title, category, copies)
VALUES
    ('Database Systems', 'Computer', 3),
    ('Digital Electronics', 'Electronics', 2),
    ('Data Structures in C', 'Computer', 2),
    ('Short Stories', 'Literature', 1);

-- Task 1
SELECT title FROM book
WHERE title LIKE 'Data%';

-- Task 2
SELECT title FROM book
WHERE title LIKE '%C';

-- Task 3
SELECT title FROM book
WHERE title LIKE '_h%';
```

7. Input:

Four supplied titles: Database Systems, Digital Electronics, Data Structures in C, Short Stories.

8. Output:

Expected output (column spacing and display format may differ in your SQL tool):

Task 1: title
Database Systems
Data Structures in C
Task 2: title
Data Structures in C
Task 3: title
Short Stories

9. Result:

The SQL statements for Experiment 5 were prepared to perform the stated tasks. On executing the setup and queries successfully, the expected results shown above should be obtained.

10. Viva-Voce Questions:

1. What is the purpose of LIKE?
2. What does % represent in a LIKE pattern?
3. What does _ represent?
4. Which pattern finds values ending in C?